CLEAN IMPLANT
FOUNDATION

The Missing Variable in Peri-implantitis

Surface Contamination, Patient Safety, and the One Decision Every Practitioner Can Make.

White Paper on Clinical, Reputational and Economic Implications for Dental Practitioners, Dental Clinics, and referring Prosthodontists.

This document summarizes evidence that sterility does not necessarily imply surface cleanliness, reviews independent findings on manufacturing/packaging residues, and outlines a structured pathway for quality assurance beyond regulatory requirements through independent verification and certification by an international non-profit organization.

CLEANIMPLANT FOUNDATION | BERLIN · NEW YORK · SHANGHAI

Chapter Summary | Four Perspectives

The Scientific Perspective

01	Peri-implantitis	Peri-implantitis affects one in five patients within five years of placement, yet prevalence has remained stable at 20% for decades despite advances in surgery, surface engineering, and maintenance therapy, pointing to an unaddressed variable.
02	Sterile Dirt	Sterility refers to the absence of viable microorganisms, whereas cleanliness refers to the absence of manufacturing residues and packaging particles. Notably, one in four sterile-packaged, ready-to-use dental implants is significantly contaminated with factory-related impurities.
03	Types of Identified Impurities	Accredited testing laboratories have identified contaminants originating from manufacturing and packaging – invisible to the naked eye, undetectable without specialized equipment, and not removed by sterilization.
04	Biological Implications of Implant Contamination	Loosely adhering particles detach during insertion and accumulate near the implant shoulder, where they trigger macrophage activation, inflammatory cytokine release, osteoclast stimulation, and progressive peri-implant inflammation.

The Manufacturers' Perspective

05	Clean Enough?	Medical device regulations require proof of sterility, mechanical stability, biocompatibility, and material composition. Biocompatibility testing addresses the toxicological aspects of extractable substances, but not the number, size, or location of individual particles. A binding threshold for particulate contamination is still missing today.

The Practitioners' Perspective

06	Clinical, Financial, and Legal Implications	Biological complications generate not only clinical burdens such as extended treatment cycles, multiple interventions, regenerative procedures, and potential implant removal, but also financial exposure (unpredictable, compounding costs), legal implications, and reputational risks.
07	Implant Choice as a Controllable Factor	Many variables influencing implant success remain beyond clinical control, but implant system selection represents a controllable factor.
08	Certification Framework	The Trusted Quality Seal certifies implant surface cleanliness beyond current regulatory requirements: a mark that cannot be purchased. It must be earned. The CleanImplant Certified Dentist/Clinic status extends this verification into the practice: making a documented commitment to quality visible to patients, referrers, and the broader professional community.

The Patients' Perspective

09	my**clean**dent.com by CleanImplant	Patients increasingly search for information online before consultation and implant treatment. my**clean**dent.com connects quality-conscious patients with CleanImplant Certified practices or clinics worldwide, creating sustained digital visibility.
10	The CleanImplant Foundation	The CleanImplant Foundation has built what did not exist before: an independent, consensus-based, and peer-reviewed global standard for implant surface cleanliness – recognized by clinicians, adapted by regulatory authorities, and carried forward by a movement of more than 200,000 dental professionals worldwide.

01 Peri-implantitis: The Persistent Clinical Challenge

A SUCCESS STORY WITH AN UNRESOLVED COMPLICATION

Dental implants represent one of the most successful treatment modalities in modern dentistry. Long-term survival rates exceeding 90% have been consistently reported across multiple clinical studies. Nevertheless, biological complications remain a significant concern.

Peri-implantitis, characterized by inflammatory bone loss around dental implants, continues to affect a substantial proportion of implant patients. Current meta-analyses suggest that approximately one in five patients develop peri-implantitis within five years of implant placement[1].

20%

of implant patients develop peri-implantitis within five years of placement.

[1] *Diaz P, Gonzalo E, Villagra LJG, Miegimolle B, Suarez MJ. What is the prevalence of peri-implantitis? A systematic review and meta-analysis. BMC Oral Health. 2022;22(1):449.*

CLINICAL PERSPECTIVE

From a clinical perspective, peri-implant disease frequently results in:

- Prolonged treatment cycles
- Repeated surgical and non-surgical interventions
- Compromised prosthetic outcomes
- Potential implant failure

ECONOMIC PERSPECTIVE

For dental practices and healthcare organizations, these complications lead to additional, economic and reputational burdens:

- Increased chair time and treatment complexity
- Higher treatment costs and resource allocation
- Patient dissatisfaction and direct attrition
- Reputational exposure within patient community

WHO'S RESPONSIBLE FOR IMPLANT FAILURE? (THE BLAME GAME)

Insufficient dental hygiene?

Patients with meticulous oral care may still develop peri-implantitis.

Surgical technique?

Yet sometimes perfectly executed placements fail.

Patients' unfavourable clinical precondition?

Quite often. However, sometimes implants placed in healthy non-smokers fail too.

Despite advances in surgical protocols, implant surface engineering, and maintenance therapy, **peri-implantitis prevalence has remained stable at approximately 20% over several decades,** pointing to a contributing variable the standard framework has not yet systematically addressed.

KEY QUESTION: *What if a fundamental variable is missing?*

02 Sterile Dirt* The Overlooked Factor

STERILITY AND CLEANLINESS

When implants are **delivered in sterile packaging** with regulatory approval, clinicians generally **assume** that the implant surface is suitable for direct placement into bone.

However, sterility and cleanliness represent two distinct concepts:
Sterility refers to the absence of viable microorganisms.
Cleanliness refers to the absence of manufacturing residues, packaging particles, or other contaminants on the implant surface.

KEY DISTINCTION:

Sterility does not predicate a clean surface.
A medical device may be sterile (microorganisms absent) while still carrying non-biological contaminants that can interact with soft- and hard-tissue during the early healing phase. These particles can lead to an uncontrolled activation of the immune system and the release of proinflammatory cytokines.

1 in 4

sterile-packaged, ready-to-use dental implants was ***significantly contaminated with factory-related impurities*** *in a quality assessment study conducted by independent testing laboratories.*

Both implants in this image collage were sterile-packaged, CE/FDA-labeled, and ready for clinical use.
Though sterile*, the above shown implant exhibits significant amounts of organic (plastic-containing) impurities..*

*) STERILE DIRT?

In 1987, four decades ago, the University Clinic Bonn, Germany, published what was likely the first quality assessment study on dental implants[1]. The authors described the findings as follows:

"All samples show particulate contaminants to varying degrees, which highlights the need for improved pre-cleaning and packaging methods. In the clinical situation, sterile dirt would be implanted along with the device..."

Nearly four decades have passed since scientists issued this call to action. It would be reasonable to expect that every implant manufacturer had long since resolved the issue. The data presented in this paper suggest otherwise.

[1] Wahl G, Tuschewitzki G-J. Contaminants on implant surfaces prior to insertion. Z Zahnärztl. Implantol 1987, III 255–260.

02 Sterile Dirt* The Overlooked Factor

AN INDEPENDENT TESTING INITIATIVE

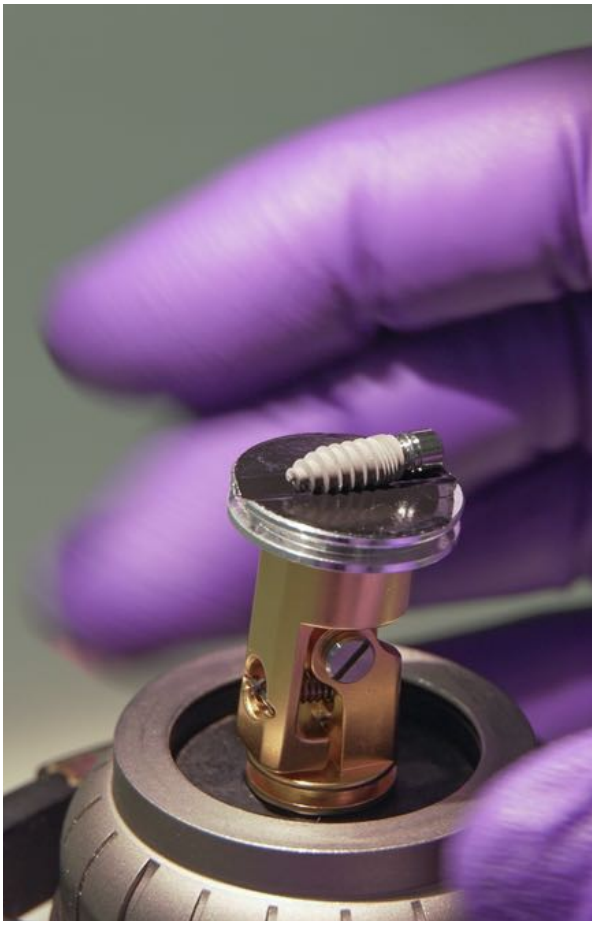
Beginning in 2016, the **CleanImplant Foundation** initiated an independent investigation of implant surface cleanliness under the direction of dentist and biologist Dr. Dirk Duddeck.

More than 450 implant systems from global manufacturers have been analyzed in subsequent quality assessment studies using standardized laboratory protocols, such as:

- Unboxing of new implant samples in an ISO Class-5 cleanroom environment
- Scanning electron microscopy (SEM)
- Energy dispersive X-ray spectroscopy (EDS)
- Time-of-flight secondary ion mass spectroscopy (ToF-SIMS)
- Cross-batch verification

The laboratories conducting these analyses operate under **ILAC MRA – DIN EN ISO/IEC 17025:2018 accreditation** to ensure analytical precision, reliability, and traceability. Technical competence and the validity of test methods are regularly re-examined by official accreditation bodies, assuring the standard of technical expertise.

In the quality assessment study conducted in 2023 on 80 implant systems from 55 manufacturers, 20 samples, i.e., **one in four implants, showed significant and clinically relevant surface contamination.** More than 50 foreign particles were detected on one-third of the implant surface.



WHAT THE STUDY DATA SHOWS

THE CLINICAL REALITY

STERILE

All implants are delivered sterile, ensuring the **absence of viable microorganisms**.
At the time of market approval, biological compatibility and cleanliness criteria play a central role.

In reality, however, one in four implant systems shows significant contamination originating from manufacturing or the packaging process.
In some cases, residues of cell-toxic cleaning agents have even been detected.

CLEANIMPLANT CONSENSUS-BASED REQUIREMENTS

STERILE AND CLEAN

Implants should not only be sterile, but should also consistently demonstrate, across batches and after market approval, the **absence of non-biological surface contaminants**.
Manufacturing residues, plastic particles, or foreign metals that can trigger an immune response at the implant-tissue interface are unacceptable.

- Cleanliness verified by SEM/EDS in accredited laboratories
- Threshold: fewer than 10 organic particles ≤50 µm per 120° viewing angle
- Zero tolerance (exclusion criterion) for foreign metals, plastics (polymers), and fluorocarbon compounds, as well as for cell-toxic chemical residues

KEY INSIGHT: *Where patient safety is of paramount importance, the Foundation's consensus-based criteria go well beyond the regulatory minimum standards for surface cleanliness.*

03 Types of Identified Impurities

INVISIBLE THREAT – VISIBLE IMPACT

Peer-reviewed reports by accredited testing laboratories have identified several categories of contaminants, some of them cell-toxic, originating from manufacturing and packaging processes. These residues are invisible to the naked eye and undetectable without specialized analytical equipment. They are not removed by sterilization.

Micrometer-scale carbonaceous particles (ranging from 0.2 µm to 7 µm in diameter) are clinically relevant, as they fall within the size range known to stimulate macrophage-mediated inflammatory responses, initiating a **cascade linked to peri-implant disease progression**. Polyethylene, for instance, has demonstrated a predictable ability to induce a pronounced inflammatory foreign body reaction, leading to bone loss[1].

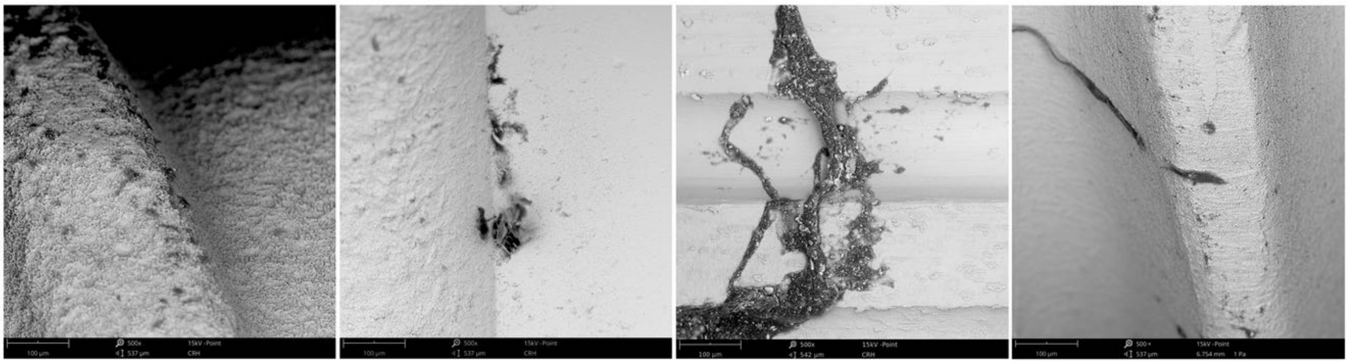
PARTICULATE CONTAMINANTS

- Polyoxymethylene (polyacetal) plastic particles
- Polysiloxane polymers
- Tungsten particles from machining tools
- Iron-chromium-nickel residues
- Copper-tin (tin bronze) particles

THIN-FILM CONTAMINANTS

- Hydrocarbon films affecting protein adsorption
- Erucamide slip agents from packaging materials
- Quaternary ammonium compounds (QACs)
- Residues of dodecylbenzene sulfonic acid (DBSA)

[1]Trindade R, Albrektsson T, Tengvall P, Wennerberg A. *Foreign Body Reaction to Biomaterials: On Mechanisms for Buildup and Breakdown of Osseointegration. Clin Implant Dent Relat Res. 2016;18(1):192-203.*



SEM images revealing significant amounts of organic (plastic-containing) particles on new, sterile-packaged dental implants

DBSA

Dodecyl-benzene sulfonic acid

WHAT MAKES THIN FILM CONTAMINANTS SO RELEVANT

DBSA, for instance, is a strong anionic surfactant and chemical irritant, **commonly used in industrial cleaning processes**, including the cleaning of medical and dental components. Residues may remain on the implant surface if cleaning and rinsing procedures are insufficient.

Due to its amphiphilic structure, DBSA **interacts with and disrupts lipid membranes**, **compromising cell integrity**[2]. DBSA has demonstrated direct cytotoxic effects, including membrane damage and reduced cell viability. Its mechanism is based on the solubilization of membrane lipids and increased permeability. From a toxicological perspective, DBSA is classified as irritating to biological tissue, underscoring its potential to interfere with cell viability upon contact[3].

[2]LArouri A, Mouritsen OG. *Membrane-perturbing effect of fatty acids and lysolipids. Prog Lipid Res. 2013;52(1):130–40.*
[3]*New Jersey Department of Health (2003). Hazardous Substance Fact Sheet: Dodecylbenzene Sulfonic Acid (CAS No. 27176-87-0). Trenton, NJ*

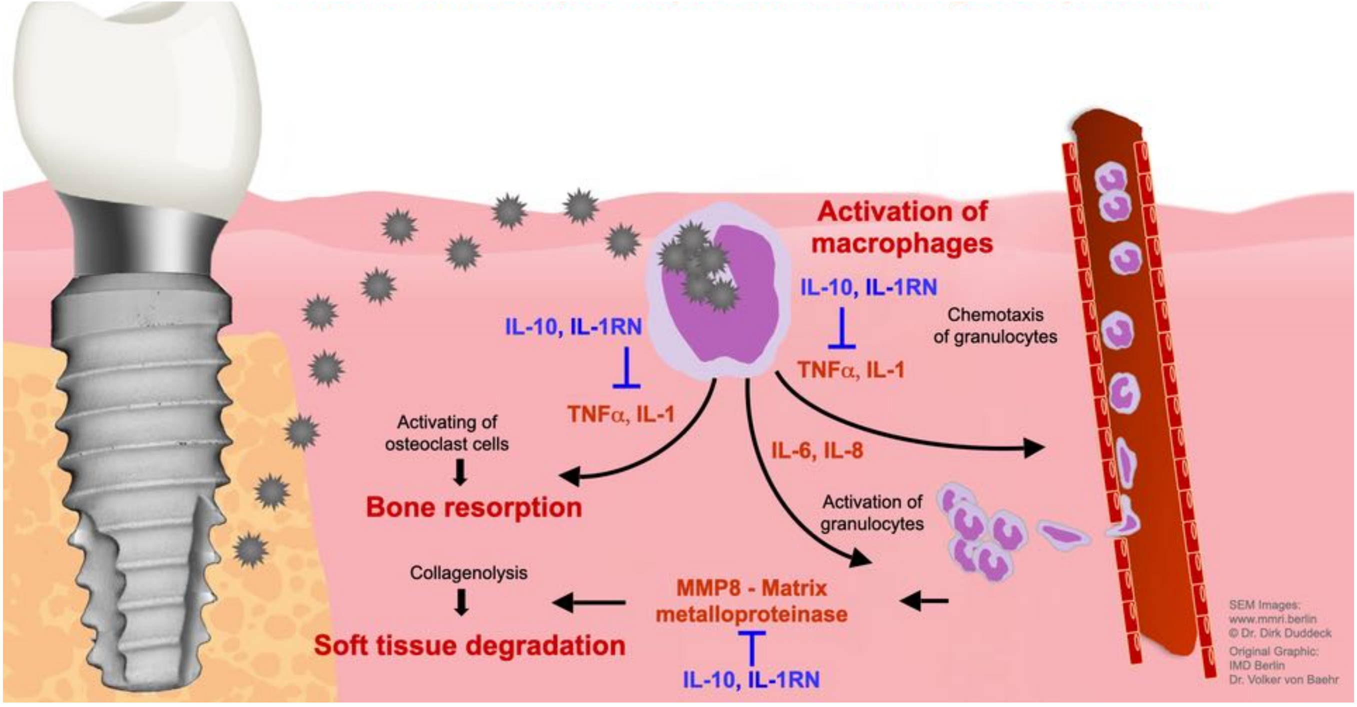
Corrosive

Acute Toxic

Irritant

Environmental Hazard

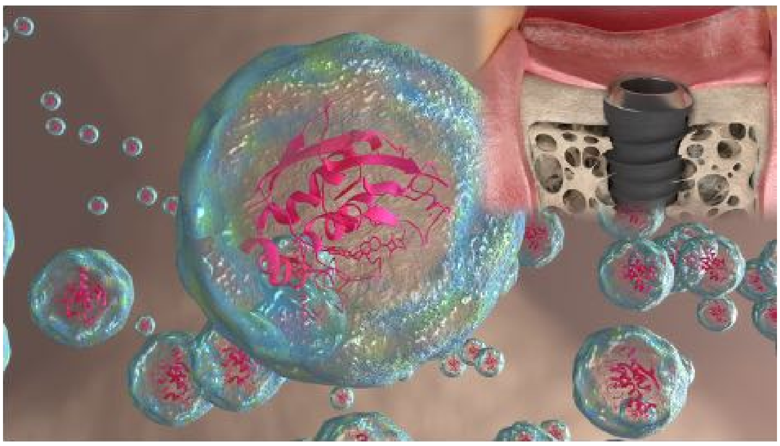
04 Biological Implications of Implant Contamination



During implant insertion, **loosely adhered particles detach and accumulate** near the implant shoulder or surrounding tissues. When foreign particles interact with host immune cells, macrophages **initiate an inflammatory cascade** involving proinflammatory cytokines such as TNF-α, IL-1, and IL-6. These mediators **stimulate osteoclast activity** and interfere with normal bone remodeling processes. Furthermore, elevated expression of matrix metalloproteinases, notably MMP-8, contributes to **soft-tissue degradation** and **progressive peri-implant inflammation**.

This often triggers the following sequence of events:

- **Early marginal bone loss and soft tissue degradation**
- **Exposure of rough implant threads**
- **Increased bacterial colonization**
- **Peri-implant mucositis**
- **Progression toward peri-implant disease**
- **Bone-loss and potential implant failure**



3D ANIMATION VIDEO

Watch the 3D animation illustrating cellular responses to particulate impurities, highlighting why these contaminants pose a biological risk.

https://www.youtu.be/JcQVHIz5UZk

KEY INSIGHT: *While contamination is not the sole cause of peri-implantitis, it may be counted among the most frequently overlooked factors compromising osseointegration. It is, moreover, one of the few factors that can already be controlled before the procedure, through the choice of a proven clean implant system.*

05 Clean Enough?

Medical device regulations worldwide prioritize patient safety by evaluating essential parameters. These standards ensure that implants introduce neither infectious nor mechanical risks to the body. Biological compatibility is also assessed: biocompatibility testing under ISO 10993 includes, among other things, determining the total content of extractable organic substances (Total Organic Carbon, TOC), a toxicologically oriented sum parameter. This test, however, answers a different question than the detailed surface cleanliness analysis performed under CleanImplant methodology, i.e., scanning electron microscopy (SEM), energy dispersive X-ray spectroscopy (EDS), and, where applicable, time-of-flight secondary ion mass spectrometry (ToF-SIMS)[1].
By its nature, the TOC determination captures neither inorganic or metallic particles nor the number, size, location, or chemical identity of individual, firmly adhering residues on the implant surface. Both methods are established, valid testing approaches for their respective purposes, and are complementary rather than interchangeable.
A systematic, standardized test for particulate surface cleanliness is, however, not currently required within most regulatory frameworks. Manufacturers are required to establish corresponding threshold values themselves, as part of their quality management system (DIN EN ISO 13485), based on a risk analysis (ISO 14971).

[1] The spatial and chemical resolution of SEM/EDS/ToF-SIMS analysis offers manufacturers a practical additional benefit: the location and precise chemical identity of a contamination substantially facilitate targeted root-cause correction in the manufacturing process, information that a purely summary TOC value cannot, by its nature, provide.

MOVING BEYOND REGULATORY BASELINES

In response to alarming study results and in the absence of corresponding ISO standards on particulate contamination of medical devices, the CleanImplant Foundation published a consensus-based guideline in September 2017. It defines, for the first time in the history of modern implantology, concrete performance criteria and contamination thresholds for dental implant surfaces. In October 2024, the U.S. Food and Drug Administration published a recommendation for performance criteria for endosseous dental implants explicitly referencing SEM/EDS surface cleanliness analysis, **the same methodology the Foundation has applied since 2017.**

While this represents a significant regulatory signal, it remains a recommendation rather than a binding requirement. **The CleanImplant Guideline remains the only framework that defines thresholds for surface cleanliness as a clinically relevant and measurable precondition for an internationally established quality mark.**

The guideline specifies that a clinically acceptable implant surface should show fewer than 10 particles, each no larger than 50 μm, within a 120-degree viewing angle from shoulder to apex, as determined under ISO Class 5 cleanroom conditions using SEM/EDS analysis in independent laboratories, officially accredited by governing bodies. These thresholds, **published in the Journal of Clinical Medicine[2]**, have been adopted by implant manufacturers worldwide. A fully revised and peer-reviewed version was released in October 2025, incorporating updated scientific evidence and refined criteria.

120°	**≤10**	**≤50 μm**	**ILAC MRA**
viewing angle, shoulder to apex	*particles per viewing angle*	*maximum particle size*	*accredited laboratories only*

[2] Duddeck DU, Albrektsson T, Wennerberg A, Larsson C, Beuer F. On the Cleanliness of Different Oral Implant Systems: A Pilot Study. J Clin Med. 2019;8(9).

Excerpt from the CleanImplant Guideline (Revision 2025)

Pictured: Dental implants undergo thorough analysis for surface cleanliness in an accredited testing laboratory

Download the latest edition of the 30-page CleanImplant Guideline here

50+ striking SEM images of implant contamination: the most comprehensive independent surface quality reference available for dental professionals and implant manufacturers

06 Clinical, Financial, and Legal Implications

Biological complications associated with implant therapy can generate cumulative effects on dental practices.

CLINICAL CONSIDERATIONS

Management of peri-implantitis frequently involves:

- Multiple surgical interventions
- Added regenerative procedures
- Potential implant removal and replacement

These treatments require **substantial chair time and resources** while often yielding uncertain outcomes.

FINANCIAL EXPOSURE

Complications generate a cascade of costs that are difficult to anticipate and harder to control:

- Unplanned retreatment pathways
- Loss of productive chair time
- Administrative burden

The cumulative **financial impact is structurally unpredictable**, and increasingly difficult to absorb at scale.

LEGAL & REPUTATIONAL RISK

Patients experiencing biological complications may express dissatisfaction through reviews or referrals, which can have an impact on a practice's or clinic's reputation.
The patients trust – in implant therapy and in the materials used – functions as a currency that moves from manufacturer to practitioner to patient – and its strength at every step depends on what can be shown, not what is assumed. Beyond reputational considerations, there is also a growing awareness of legal aspects: as scientific discussion around implant surface quality continues to evolve. Practices and clinics may increasingly be expected to demonstrate structured and well-documented implant selection criteria.

> ***For multi-site organizations (DSOs)**, the risk profile is multiplied across volume: procurement decisions and clinical standardization influence complication exposure and brand reputation.*

07 Implant Choice as a Controllable Factor

Many variables influencing implant success remain beyond clinical control.
However, implant system selection represents a variable that clinicians and clinics can actively manage. Incorporating independently **verified quality data** and cleanliness information into the decision process for a suitable implant system enables practices to add an additional layer of quality assurance in implant therapy.

Not all variables that determine implant outcomes fall within the reach of practice governance, but identifying which ones do is the starting point for **meaningful quality control**.

WITHIN
YOUR SCOPE OF CONTROL

- **Implant system selection**
- Procurement and inventory controls
- Documentation of quality assurance criteria
- Team education and treatment protocols
- Patient communication

OUTSIDE
YOUR SCOPE OF CONTROL

- Systemic health / medications
- Patient hygiene adherence
- Genetic inflammatory predisposition
- Long-term implant maintenance compliance
- Reputation

Implant choice is different in kind. **It is the one variable in this list that is determined entirely at the clinical decision point,** before the patient is in the chair, before any treatment has started, and before osseointegration has begun. Incorporating independently verified cleanliness data into implant selection is therefore not an additional quality layer but **the most direct and actionable intervention available to reduce the risk** of contamination-related early treatment complications such as peri-implant mucositis, loss of attachment, significant bone resorption, and peri-implantitis.

Pictured: Dental implants mounted on special sample holders for thorough scanning-electron microscopic analysis

KEY INSIGHT: *Independent verification supports a convincing narrative – Implant choice based on this cross-batch verified cleanliness and clinically documented survival rates, rather than sole reliance on marketing claims or price tiers.*

08 The Certification Framework Part 1: The Trusted Quality Seal

THE TRUSTED QUALITY SEAL

The Trusted Quality Seal is an independent quality mark awarded by the CleanImplant Foundation to implant systems that meet consensus-based surface cleanliness criteria exceeding current regulatory requirements. Regulatory conformity assessment – commonly referred to as regulatory approval – confirms sterility, tested cleanliness criteria, and mechanical safety at the time of certification[1]. Independent sampling by the CleanImplant Foundation of implant samples already on the market shows that this level of cleanliness is not consistently maintained by numerous implant manufacturers. The Trusted Quality Seal goes further: it certifies that, across lots, the implant surface is – within the limits of analytical detection – free of manufacturing residues, particulate contamination, and chemical deposits. The seal must be renewed every two years. It cannot be purchased or self-declared.

[1] Under the EU Medical Device Regulation (MDR (EU) 2017/745), market access is granted through a conformity assessment procedure – typically involving a Notified Body – rather than a government approval in the sense familiar from other regulatory systems (e.g., FDA clearance/approval in the United States). This assessment is based on the technical documentation and test samples submitted at the time of certification. Ongoing compliance with these criteria during serial production is the manufacturer's responsibility under the post-market surveillance obligations of Art. 83–86 MDR, and is not ensured through repeated regulatory testing of every individual lot.

How certification works:
Before surface cleanliness is tested, a system must demonstrate a clinical survival rate of at least 95% for 2 years, supported by peer-reviewed publications or post-market clinical follow-up documentation. Samples from different batches are acquired anonymously through mystery shopping, ensuring that what is tested reflects what actually reaches the clinical environment. All samples are analyzed by accredited laboratories (ILAC MRA – DIN EN ISO/IEC 17025:2018) using SEM, EDS, and, where indicated, ToF-SIMS for molecular-level characterization. Every result is independently peer-reviewed by the CleanImplant Scientific Advisory Board before any certification decision is made.

KEY INSIGHT: *For the clinician, the Trusted Quality Seal transforms implant choice into a communicable quality commitment: the ability to document and communicate a clinical decision whose quality has been repeatedly and independently verified, even years after market approval, according to one of the highest available standards.*

The CleanImplant Foundation invites every manufacturer to join the recurring quality assessment studies

Of the **more than 450 implant systems assessed** since 2016, a meaningful and growing share have consistently met the cleanliness threshold across multiple production batches, demonstrating that manufacturing that is contamination-free within the limits of analytical detection is not aspirational, but operational. The data reveal no systematic relationship between cleanliness performance and price tier, country of manufacture, or market position. Where contamination occurs, it reflects specific production and packaging decisions, not the inherent limitations of any category of manufacturer.

WE AWARD EXCELLENCE.

CLEAN IMPLANT FOUNDATION

Implant manufacturers/brands whose implants have met the required thresholds at least once in the past:

Across every major market segment, manufacturers have demonstrated that near-contamination-free production is achievable. The following manufacturers have met the Foundation's consensus-based cleanliness criteria and required survival rates with one or more implant types at least once in the past*:

ANTHOGYR	BTI BIOTECHNOLOGY INSTITUTE	IZEN IMPLANT	RITTER IMPLANTS
ASTRA TECH	CHAMPIONS IMPLANTS	LYRA ETK	SLH
BIOTECH DENTAL	DENTIS	MEDENTIS MEDICAL	SOUTHERN IMPLANTS
BIOTEM	DENTIUM	NOBEL BIOCARE	STRAUMANN
BREDENT MEDICAL	GLOBAL D	NUCLEOSS	SWISS DENTAL SOLUTIONS

This list makes no claim to completeness and includes only implant systems that have met the criteria in the past. The absence of a manufacturer from this list permits no conclusion about its product quality; it merely means that no active, successfully completed certification currently exists, or that the manufacturer has not participated in the process.

*) The complete list of currently certified systems is publicly accessible at: ***cleanimplant.com/clean-implants/***

08 The Certification Framework
Part 2: Certified Dentist & Certified Clinic

Certification as a CleanImplant Certified Dentist or Clinic

A commitment to using independently verified implant systems is a clinical decision of the highest standard. CleanImplant certification formalizes that commitment, making it visible, documented, and communicable to the patients and referring colleagues who value it. The certification program is not open to all. It is available exclusively to practices and clinics that use an implant system carrying the coveted Trusted Quality Seal.

The exclusive certification as a CleanImplant Certified Dentist/Clinic provides:

CLINICAL QUALITY & DOCUMENTATION

- Peer-reviewed surface analysis data for implant selection and patient consultations
- A documented QA framework translating independent analytical data into an auditable clinical standard
- Patient education materials distinguishing sterility from surface cleanliness
- The trust of independent verification, carried directly into patient relationships

PRACTICE AND CLINIC IDENTITY & COMMUNICATION

- Personalized quality certificate and certification logo
- Patient-facing materials: reception displays, waiting room brochures, stickers for estimates and correspondence
- Social media templates and marketing support for consistent quality messaging
- A tailored practice profile and managed landing page on mycleandent.com

PROFESSIONAL DEVELOPMENT & NETWORK

- Recognition within a global community of leading implantologists
- Sustained digital visibility through the Foundation's professional network, social media, newsletter, and international conference reach
- CME-accredited continuing education and exclusive training courses
- A dedicated personal contact throughout certification and beyond

> ***KEY INSIGHT:*** *Certification as a CleanImplant Certified Dentist/Clinic is more than a quality recognition. It is the entry into an exclusive international community of practitioners who consistently prioritise implant quality.*
> *Membership documents a verified and visible commitment to patients, peers, and the profession.*

The **core benefit of practice certification** is a individual **practice profile on mycleandent.com.** On the following pages we introduce the patient-facing platform with a searchable directory of certified practices. It connects quality-conscious patients worldwide with certified practices and clinics. The complimentary landing page on mycleandent.com is the most significant benefit of certification: a platform that translates clinical commitment and **quality standards** into **visibility for new implant patients** – at the precise moment they are making their decision.

09 mycleandent.com
Visibility, Communication and Patient Reach

Patients approaching implant treatment increasingly research their options before a consultation takes place. They search by location, by specialty, by quality signals they can understand and trust. **mycleandent.com** is where that search leads to a CleanImplant Certified practice or clinic.

mycleandent

by CleanImplant

> **Every dentist and clinic listed on mycleandent.com uses an implant system that carries the Trusted Quality Seal**, independently verified by the CleanImplant Foundation to be not only sterile but also free of detectable surface contamination. For patients who ask this question, mycleandent.com is where the answer is documented, searchable, and linked to a specific practice or clinic.

Rather than functioning as a static directory, the platform actively manages digital visibility on behalf of each certified practice or clinic through three operational pillars:

LOCAL PRESENCE

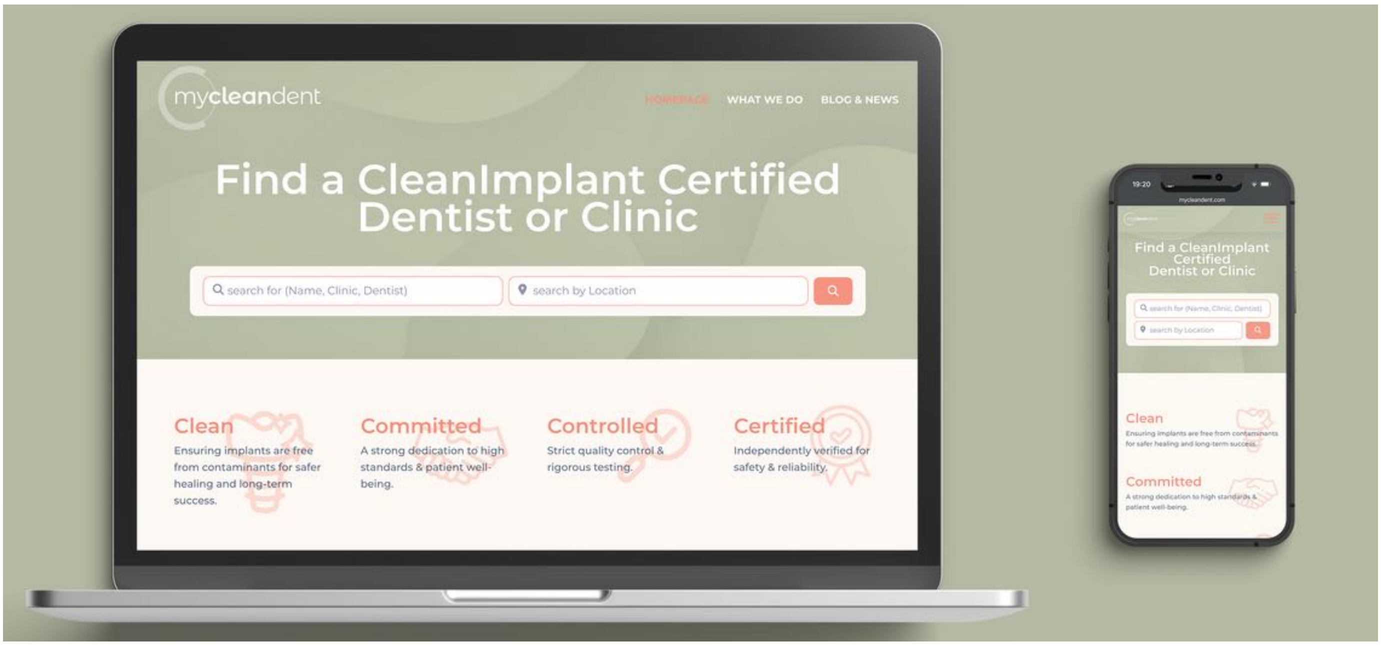
Each profile is supported by **Google Business Profile optimization** and targeted campaigns at portal level, helping ensure that patients with active treatment intent in the region find the certified practice or clinic at the right moment.

TRUST THROUGH CONTENT

Professionally conducted **expert interviews and dedicated landing pages** built around each practice or clinic's specializations give patients substantive context before the first consultation, arriving informed and aligned with the quality standard the practice or clinic represents.

TECHNICAL OPTIMIZATION

Structured data implementation supports **precise classification** by specialty, location, and patient reviews, **improving organic search visibility**. As patient search behavior increasingly extends to AI-assisted discovery tools, well-structured portal profiles are better positioned to be found.



Pictured: The mycleandent directory for quality conscious patients looking for certified practitioners

09 mycleandent.com Visibility, Communication, and Patient Reach

How mycleandent.com extends the reach of every CleanImplant-certified practice and clinic:

A presence on mycleandent.com **adds a layer of visibility** that operates independently of and in addition to any existing digital presence a practice or clinic maintains. As a platform dedicated exclusively to independently verified implant care, aggregating certified practices and clinics across the globe, it **carries thematic authority** that search engines classify as specialist expertise in this field and rank accordingly. The result is a reach into patient search behaviour that a single-location website, by its nature, cannot achieve on its own.

This reach translates into several concrete structural advantages for every listed practice or clinic:

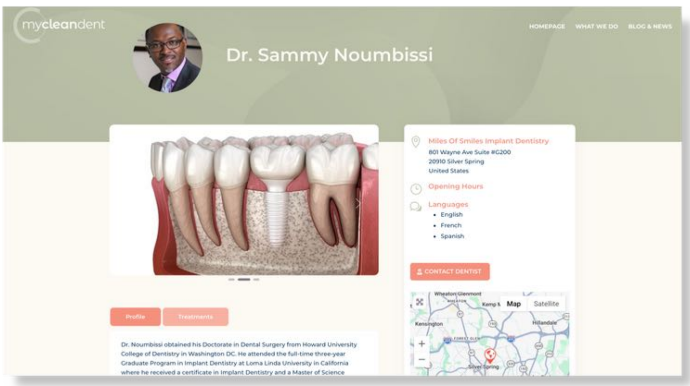
Search visibility: The platform's authority in this specialty area reflects positively on the search visibility of every associated practice or clinic, generating a sustained benefit that grows as the network expands.

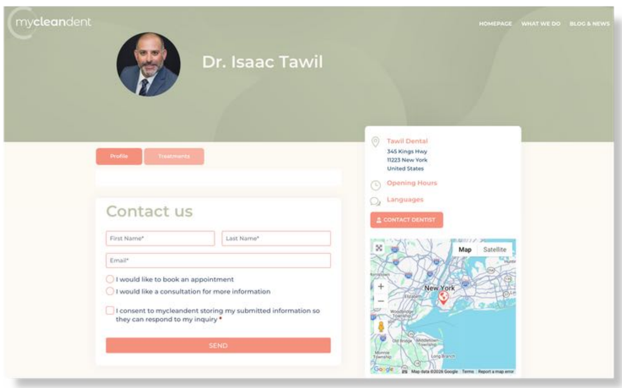
Early-stage patient capture: mycleandent.com addresses the questions patients ask before they have chosen where to go: about costs, materials, healing timelines, and risk. These patients arrive at a certified practice or clinic already informed, and already aligned with the quality standard it represents, a qualitatively different starting point for the consultation.

Referrer confidence: Certification communicates a documented quality commitment not only to patients but also to referring colleagues. In a professional environment where implant outcomes reflect on the referring prosthodontist, as much as the treating surgeon, independently verified quality is a credible signal that strengthens professional relationships and referral networks.

AI-assisted discovery: As patients increasingly use AI-assisted tools to find specialist care, structured portal profiles with independently verified quality credentials are better positioned to appear in location-based recommendations than individual practice or clinic websites.

Community and network reach: The CleanImplant Foundation's community of more than 200,000 dental professionals across social media, its newsletter, and international conference presence helps ensure that certified practices and clinics benefit from the visibility of the broader movement: a network effect that no individual practice or clinic could build independently.





Pictured: Exemplary landing pages for dentists and clinics listed on my**clean**dent.com

KEY INSIGHT: *The cumulative effect is sustainable visibility, continuously maintained, continuously compounding, directed toward patients who are actively seeking exactly what a certified practice or clinic represents. It is included at no additional cost as an integral part of the CleanImplant Certified Dentist/Clinic membership.*

10 The CleanImplant Foundation: Scientific, Independent, Peer-reviewed

Who is the CleanImplant Foundation?

The CleanImplant Foundation was established in Berlin in 2016 by Dr. Dirk U. Duddeck, dentist and biologist, in response to a structural gap in the dental implant market: no independent body existed to systematically assess **whether commercially available implant systems were clean, not merely sterile**, at the point of clinical use.

Since then, the Foundation has spent years building an internationally recognized reputation by having the surface cleanliness of dental implants systematically tested worldwide and, independently of any manufacturer influence, certified. All analyses are commissioned through accredited partner laboratories operating under ILAC MRA – DIN EN ISO/IEC 17025:2018 accreditation. **Results are peer-reviewed by an internationally recognized Scientific Advisory Board**, whose members define the cleanliness criteria, validate the methodology, and independently review every certification decision, before any finding is published or any certification is made.

The CleanImplant Foundation's mission is carried forward by a global network of **more than 50 Key Opinion Leaders and Ambassadors from over 20 countries**: clinicians and researchers who present the study results at international conferences, publish peer-reviewed papers, and advocate for verified implant quality in their regions.

The Foundation's work is not adversarial. Its purpose is not to indict manufacturers; it is to provide clinicians and practices with independently verified data to make informed decisions about the systems they place.

Excellence deserves recognition. The Trusted Quality Seal is how that recognition is formalized.

Pictured: The CleanImplant Foundation’s Board and global ambassadors at the biannual CleanImplant Summit in Berlin

The Foundation's Scientific Impact and Global Reach

What began as a single concern - whether commercially available implants were truly clean at the point of clinical use - has **grown into a global movement for better quality in implant dentistry**.

Since 2016, the CleanImplant Foundation has analyzed more than 450 implant systems across every major global market, built a **community of over 200.000 dentists and industry affiliated professionals**, and established a network of certified practices around the globe.

Further research into the clinical relevance of implant surface contamination is actively promoted and commissioned in collaboration with **renowned Universities worldwide**, including Charité University Medicine Berlin, Germany, the Sahlgrenska Academy, University of Gothenburg, Sweden, and the University of Zurich, Switzerland. The Foundations’ research has been published in peer-reviewed journals and presented at leading international congresses.

Find more information: **www.cleanimplant.org**

*“**We owe every patient the highest quality products possible**. The situation on the market shows the urgent need for independent, not industry-driven, quality assessments. Safety and well-being of our patients always come first!”*

Dr. Dirk U. Duddeck
Founder and Managing Director

*”The reason I am involved: we have to expect that the dental devices have attained a certain quality of cleanliness and surface technologies. **This is what the quality assessments of CleanImplant bring to this industry**. And I welcome it!”*

Dr. Michael Norton
Scientific Advisory Board

*”**Once you see it, you cannot unsee it!** The work and fight of the CleanImplant Foundation helps to understand. One day, immunologists will wonder how they overlooked this important detail of implant contamination”*

Dr. Miguel Stanley
Ambassador Portugal

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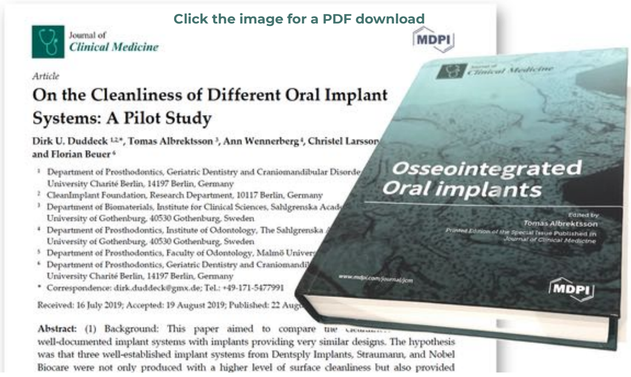

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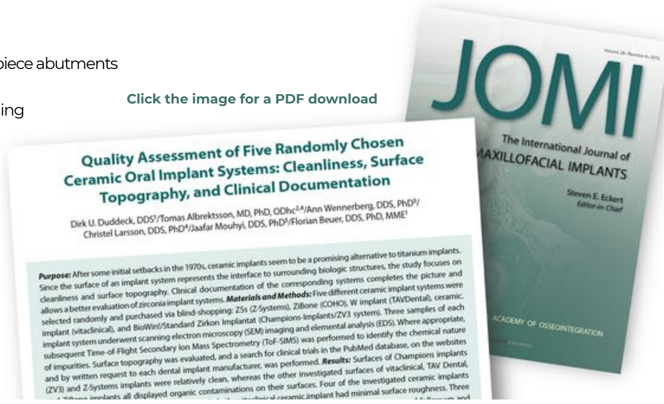

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The equilibrium between an implant surface and surrounding host tissue is fragile, particularly in the early healing phase. *Any factor* that disrupts it at the moment of placement – before osseointegration has established sufficient stability – can initiate an inflammatory cascade with lasting consequences.

The surface condition of the implant itself is one such factor.

Disclosure:
This white paper is educational in nature. It synthesizes peer-reviewed literature, publicly available descriptions of the certification for dental practitioners, dental clinics, and DSOs, the testing processes, and performance criteria for the Trusted Quality Mark. It does not replace clinical judgement, manufacturer instructions, or local regulatory requirements.

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